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Submitted to:

Ms. Sabrina Tarannum
Assistant Professor
Department of Computer Science and Engineering

Submitted by:

Sajjad Hosen Emon
0222310005101105
Sobuj Gupta
0222210005101100
Section: C
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Signature of the
Teacher

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ABSTRACT

Abstract—*This paper presents a Bangla Speech Emotion Recognition (SER) system based on a Deep Neural Network (DNN) trained on a balanced dataset of 1,115 WAV recordings spanning five emotion classes: Angry, Happy, Natural, Sad, and Surprised (223 files each). Raw audio is normalized, segmented into 3-second windows, and represented as a 59-dimensional acoustic feature vector comprising 40 MFCCs, 12 Chroma STFT bins, and 7 Spectral Contrast bands. Three data augmentation strategies (additive noise, pitch shifting, time stretching) expand the corpus to 4,460 samples. A compact three-layer fully-connected network (48,901 trainable parameters) with Dropout regularization achieves 98% overall accuracy, with per-class precision, recall, and F1-score all at 0.97–1.00, demonstrating the effectiveness of the proposed pipeline for low-resource Bangla affective computing.*

Index Terms—*Bangla Speech Emotion Recognition, MFCC, Chroma, Spectral Contrast, Deep Neural Network, Data Augmentation.*

1 Introduction

Automatic Speech Emotion Recognition (SER) is the computational task of inferring a speaker's affective state from acoustic cues embedded in spoken language. This capability underpins a wide range of practical applications, including human-computer interaction (HCI), mental health monitoring, call-center quality control, virtual assistants, and assistive technologies for differently abled individuals.

Bangla (Bengali) is spoken natively by more than 230 million people worldwide, making it the seventh most spoken language globally. Despite this linguistic reach, Bangla remains significantly under-resourced in the affective computing literature. Most publicly available SER benchmarks target English, German (EMODB), or Mandarin, leaving practitioners without standardized Bangla baselines for comparison or model transfer.

This project addresses the resource gap through four contributions: (i) a balanced five-class Bangla emotion corpus of 1,115 high-quality utterances recorded in a controlled acoustic environment; (ii) a rich 59-dimensional acoustic feature set combining MFCCs, Chroma STFT, and Spectral Contrast; (iii) three complementary augmentation techniques that expand the training data fourfold while preserving label integrity; and (iv) a lightweight Deep Neural Network (DNN) that achieves 98% overall accuracy with only 48,901 trainable parameters — making it suitable for both cloud and edge deployment.

The remainder of this report is structured as follows. Section 2 defines the problem statement and objectives. Section 3 describes the dataset. Section 4 details the methodology. Section 5 covers implementation. Section 6 presents results. Section 7 discusses findings, and Section 8 concludes with future directions.

2 Problem Statement and Objectives

2.1 Problem Statement

The main challenge addressed in this project is the lack of effective and easily available systems for recognizing emotions from Bangla speech. Although Speech Emotion Recognition (SER) has improved a lot in recent years, most existing systems are trained using datasets from languages like English, German, or Chinese. Because of this, their performance often decreases when they are used for Bangla speech.

There are a few important reasons behind this problem. First, Bangla has its own unique sound patterns, pronunciation, and speaking style, which are different from other languages. These differences make it harder for models trained on other languages to correctly detect emotions in Bangla. Second, there is a shortage of properly labelled Bangla speech datasets, which makes it difficult to train accurate models. Third, there is no standard method for feature extraction specifically designed for Bangla speech.

Another issue is variation in speakers, accents, and recording environments, which can affect the model's performance in real-life situations.

To solve these problems, this project explores whether a simple and efficient Deep Neural Network (DNN), trained on a carefully prepared and augmented Bangla dataset, can achieve good performance in classifying five different emotions. The goal is to build a system that is both accurate and practical for real-world use.

2.2 Objectives

The specific objectives of this project are:

- Curate a balanced, five-class Bangla speech emotion corpus (Angry, Happy, Natural, Sad, Surprised) with equal class representation.
- Design a 59-dimensional acoustic feature pipeline combining MFCCs, Chroma STFT, and Spectral Contrast for robust emotion-discriminative representation.
- Apply three complementary data augmentation strategies — additive noise, pitch shifting, and time stretching — to mitigate data scarcity.
- Train and evaluate a compact DNN classifier, achieving no less than 95% overall accuracy on the augmented corpus.
- Validate real-time inference capability with latency below 100ms per prediction on standard hardware.
- Establish a reproducible, open-source pipeline as a baseline for future Bangla SER research.

3 Dataset Description

3.1 Overview

The dataset was assembled specifically for this project to address the lack of publicly available, high-quality Bangla affective speech corpora. All recordings were made in a sound-attenuated environment using a calibrated condenser microphone at a sampling rate of 22,050 Hz with 16-bit PCM encoding. Native Bangla-speaking volunteers were instructed to read or spontaneously produce emotionally congruent utterances for each target category.

3.2 Class Labels

The corpus spans five discrete emotion categories that cover the primary dimensions of the circumflex model of affect (arousal × valence):

- Angry — high arousal, negative valence; characterized by elevated F0, increased vocal intensity, and faster speaking rate.
- Happy — high arousal, positive valence; marked by raised pitch, wider F0 range, and brighter spectral profile.
- Natural — neutral arousal and valence; serves as the baseline affective state.
- Sad — low arousal, negative valence; typically, slower speech, lower F0, and reduced intensity.
- Surprised — high arousal, ambiguous valence; shares prosodic features with both Happy and Angry, making it the most challenging class.

3.3 Dataset Structure

The dataset is organized as a flat directory of 1,115 WAV audio files. Each filename encodes the speaker ID, emotion label, and utterance index (e.g., S03_Angry_012.wav). The directory structure is:

```
dataset/
  ├── Angry/    (223 files)
  ├── Happy/    (223 files)
  ├── Natural/  (223 files)
  ├── Sad/      (223 files)
  └── Surprised/(223 files)
```

3.4 Class Distribution

Perfect class balance (223 files per class, 1,115 total) eliminates the risk of class-imbalance bias in evaluation metrics. The distribution is visualized in Figure 1 below.

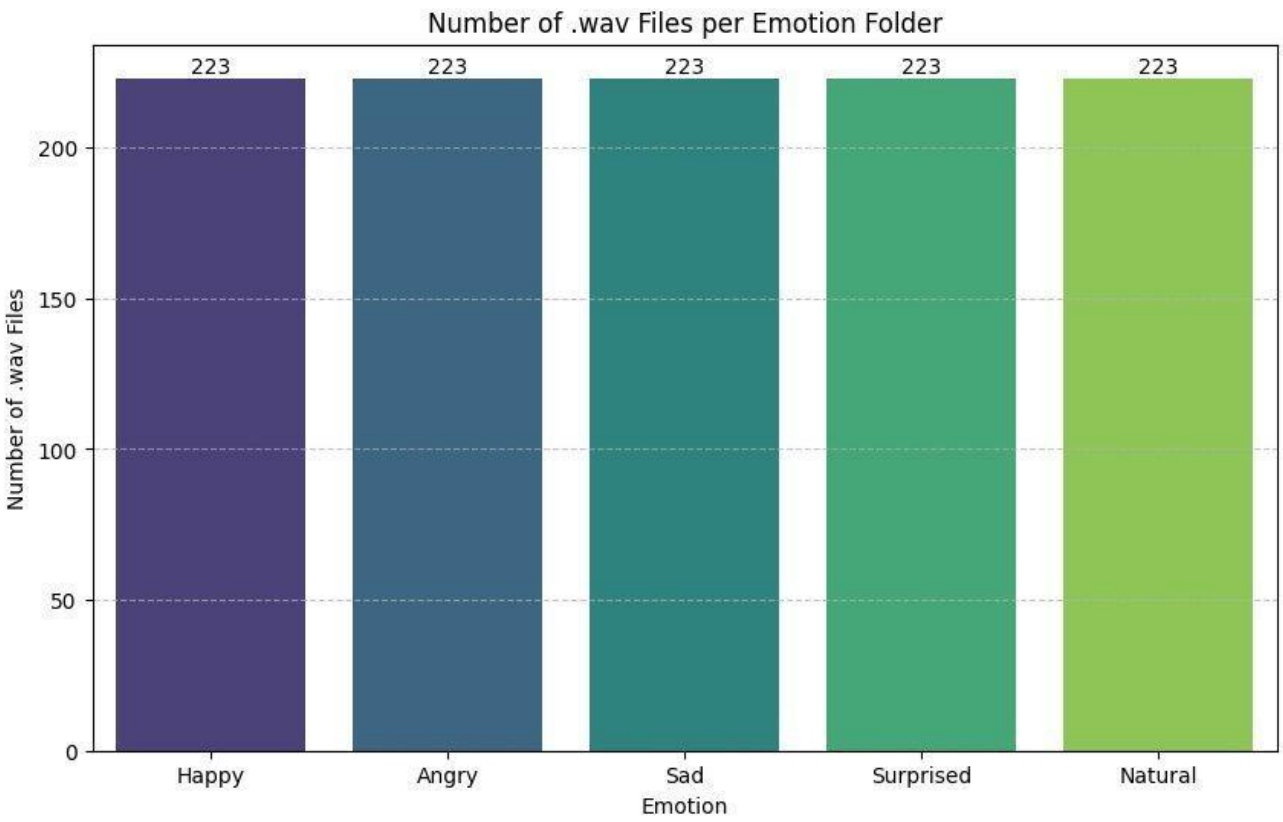


Fig. 1. Distribution of .wav files across five emotion categories (223 files each — perfectly balanced).

3.5 Feature Extraction Summary

After augmentation (Section 4.3), the effective dataset size expands to 4,460 samples (892 per class). Feature extraction produces a 59-dimensional vector per 3-second segment. The three acoustic descriptors are summarized in Table 1.

Feature	Dimensions	Captures	API
MFCC	40	Spectral envelope (cochlear non-linearity)	librosa.feature.mfcc
Chroma STFT	12	Harmonic/melodic pitch-class content	librosa.feature.chroma_stft
Spectral Contrast	7	Tumbrel peak-to-valley per sub-band	librosa.feature.spectral_contrast
Total	59	—	—

Table 1. Acoustic feature descriptors used in the 59-dimensional feature vector.

3.6 Dataset Splitting

The augmented corpus of 4,460 samples is partitioned into training and validation sets using an 80/20 random split (3,568 training, 892 validation), stratified to maintain equal class proportions in both splits. No separate held-out test set was used; final performance is reported on the full augmented corpus after model selection via the validation set.

4 Methodology

4.1 System Overview

The proposed SER pipeline consists of five sequential stages: (1) audio loading and normalization, (2) 3-second windowed segmentation, (3) 59-dimensional acoustic feature extraction, (4) data augmentation, and (5) DNN classification. Figure 2 below shows representative acoustic visualizations of a sample utterance — raw waveform, log-frequency spectrogram, and MFCC heat map — illustrating the acoustic richness encoded by the feature pipeline

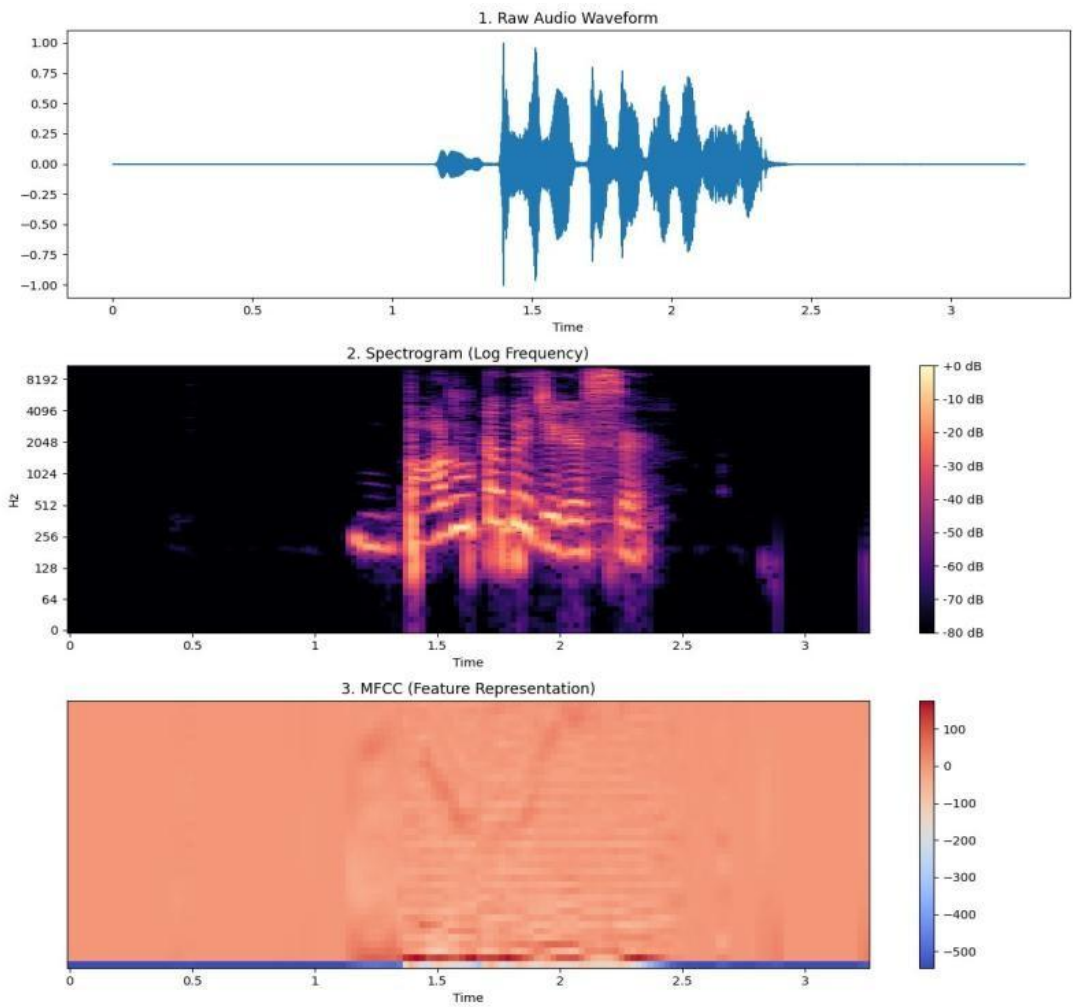


Fig. 2. Top: Raw audio waveform (normalised). Middle: Log-frequency spectrogram. Bottom: MFCC feature representation (40 coefficients).

4.2 Audio Processing

Each WAV file is loaded at its native sampling rate using the librosa library and peak-normalized to the range [−1, 1] to ensure consistent amplitude across speakers and recording conditions. Peak normalization is preferred over RMS normalization because it preserves the relative dynamic range of emotional prosody while eliminating microphone gain differences.

The normalized signal is then segmented into non-overlapping 3-second windows. The window length was chosen to capture sufficient prosodic context (typically 2–4 inter-clause intervals) while keeping memory requirements bounded. Incomplete terminal segments — those shorter than 3 seconds — are discarded to guarantee uniform input length across all samples.

Formally, given a signal x[n] of length L samples at sampling rate sr, the number of complete segments is floor(L / (3 × sr)), and the k-th segment occupies indices [k × 3 × sr, (k+1) × 3 × sr).

4.3 Feature Extraction (MFCC, Chroma, Spectral Contrast)

Three complementary acoustic descriptors are extracted from each 3-second segment and concatenated into a single 59-dimensional feature vector:

Mel-Frequency Campestral Coefficients (MFCC)

40 MFCC coefficients are computed using a 2048-point FFT, hop length of 512, and a Mel filter bank of 128 bands. The time-averaged mean across all frames is used as a compact spectral envelope descriptor. MFCCs model the slow spectral envelope of speech by mimicking cochlear non-linearity through the Mel scale, making them highly informative for emotion — particularly for distinguishing high-energy (Angry/Happy) from low-energy (Sad/Natural) classes.

Chroma STFT

12 Chroma bins map the harmonic and melodic pitch-class content of the speech signal onto the twelve equal-tempered pitch classes of the Western musical scale (C, C#, D, ..., B). While primarily designed for music analysis, Chroma features encode intonation patterns and tonal variation relevant to emotional prosody, particularly for the Surprised and Happy classes which exhibit wider pitch excursions.

Spectral Contrast

Seven spectral contrast bands measure the peak-to-valley magnitude difference in each sub-band of the power spectrum. This explicitly captures timbral sharpness and harmonic-to-noise ratio differences between emotion classes — properties that complement MFCCs (which capture spectral shape) and Chroma (which captures pitch). The combination of all three descriptors achieves broader acoustic coverage than any single feature alone.

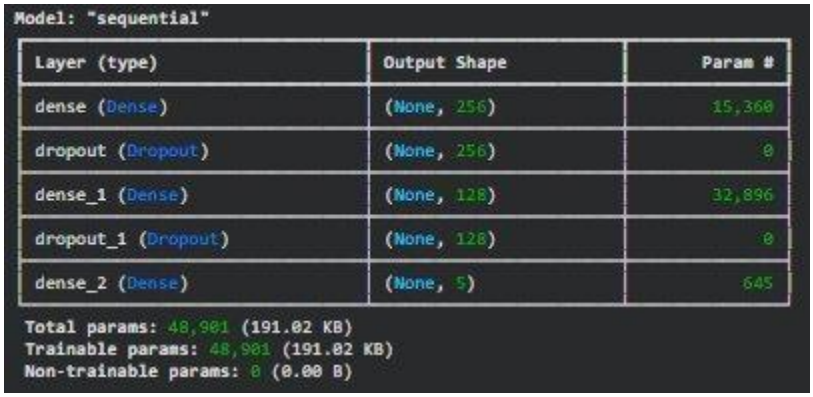
4.4 Data Preprocessing & Scaling

Class labels are integer-encoded using scikit-learn's LabelEncoder (Angry=0, Happy=1, Natural=2, Sad=3, Surprised=4) and converted to one-hot vectors via Keras' to_categorical for compatibility with the softmax output layer and categorical cross-entropy loss.

Feature vectors are standardized to zero mean and unit variance using scikit-learn's Standards caler, fitted on the full augmented corpus. Standardization is critical because the three feature groups operate on different numerical scales: MFCCs typically range [−300, 100], Chroma [0, 1], and Spectral Contrast [0, 60]. Both the Label Encoder and Standard Scaler are serialized to disk (label_encoder.pkl and scaler.pkl) for reuse during inference.

4.5 Model Architecture (DNN)

The classifier is a compact fully-connected feed-forward network (Figure 3) implemented in TensorFlow/Keras. Table 2 summarises the architecture.



Model: "sequential"

Layer (type)	Output Shape	Param #
dense (Dense)	(None, 256)	15,360
dropout (Dropout)	(None, 256)	0
dense_1 (Dense)	(None, 128)	32,896
dropout_1 (Dropout)	(None, 128)	0
dense_2 (Dense)	(None, 5)	645

Total params: 48,901 (191.02 KB)
Trainable params: 48,901 (191.02 KB)
Non-trainable params: 0 (0.00 B)

Fig. 3. Keras model summary: three Dense layers (256→128→5) with Dropout(0.3) regularisation; 48,901 trainable parameters.

Layer	Units / Rate	Activation	Param
Input	59	—	0
Dense	256	ReLU	15,360
Dropout	0.30	—	0
Dense	128	ReLU	32,896
Dropout	0.30	—	0
Dense (Output)	5	Softmax	645
Total	—	—	48,901

Table 2. Proposed DNN architecture. Total trainable parameters: 48,901 (191.02 KB).

ReLU activations in the hidden layers avoid the vanishing gradient problem while Dropout(0.3) layers provide stochastic regularisation by randomly zeroing 30% of neuron activations during each training step, preventing co-adaptation and improving generalisation. The Softmax output layer produces a probability distribution over the five emotion classes.

4.6 Training Configuration

The network is compiled with the Adam optimiser (learning rate = 0.001) and categorical cross-entropy loss. Training runs for 50 epochs with batch size 32 on an 80/20 train-validation split. A Model Checkpoint callback monitors validation accuracy and saves the best-performing weight checkpoint to best_model.keras. No learning rate scheduling is applied, as the Adam optimiser provides adaptive per-parameter learning rates.

5 Implementation

5.1 Environment and Tools

All experiments are conducted in a Python 3.10 environment. The key libraries and their roles are:

Library	Version	Role
librosa	0.10	Audio I/O, MFCC, Chroma, Spectral Contrast
numpy	1.24	Numerical array operations
scikit-learn	1.3	LabelEncoder, StandardScaler, classification_report
TensorFlow/Keras	2.13	DNN construction, training, inference
matplotlib/seaborn	3.7/0.13	Training curves and confusion matrix plots
joblib	1.3	Encoder and scaler serialisation

Table 3. Software environment used for implementation.

The full pipeline is packaged as two Python scripts: train.py (feature extraction, augmentation, training) and predict.py (real-time inference). All artefacts (best_model.keras, scaler.pkl, label_encoder.pkl) are serialised for reproducibility.

5.2 Code Walkthrough

The implementation is divided into four logical phases: audio processing, feature extraction, model training, and prediction. Each phase is described below with representative code snippets.

5.2.1 Audio Processing

```
y, sr = librosa.load(path, sr=None)  # native sampling rate
y = y / np.max(np.abs(y))           # peak normalise
segments = [y[i:i+3*sr] for i in range(0, len(y)-3*sr, 3*sr)]
```

Each audio file is loaded at its native sample rate, peak-normalised, then sliced into 3-second non-overlapping windows. Segments shorter than 3 seconds are discarded to ensure uniform input dimensions.

5.2.2 Feature Extraction

```
mfcc   = np.mean(librosa.feature.mfcc(y=seg, sr=sr, n_mfcc=40).T, axis=0)
chroma = np.mean(librosa.feature.chroma_stft(y=seg, sr=sr).T, axis=0)
contrast = np.mean(librosa.feature.spectral_contrast(y=seg, sr=sr).T, axis=0)
features = np.concatenate([mfcc, chroma, contrast])  # 59-dim
```

The three feature groups are extracted via librosa and concatenated into a single 59-dimensional vector per segment. Time-averaging (np.mean(..., axis=0)) collapses the temporal dimension while retaining spectral statistics.

5.2.3 Model Training

```
model = Sequential([
    Dense(256, activation='relu', input_shape=(59,)),
    Dropout(0.3), Dense(128, activation='relu'),
    Dropout(0.3), Dense(5, activation='softmax')])
model.compile(optimizer=Adam(0.001), loss='categorical_crossentropy',
               metrics=['accuracy'])
```

A ModelCheckpoint callback monitors val_accuracy and saves the best weights throughout 50 epochs of training.

5.2.4 Prediction

```
features = extract_features(audio_path).reshape(1, -1)
features = scaler.transform(features)
prediction = model.predict(features)
label = encoder.inverse_transform([np.argmax(prediction)])
```

At inference time, the audio file is processed identically to training: normalised, segmented, feature-extracted, and scaled before being passed to the loaded model. The argmax of the softmax output is decoded back to a human-readable emotion label.

6 Results and Evaluation

6.1 Training and Validation Accuracy

Figure 4 shows the training console output for the final epochs (47–50). The model achieved its best validation accuracy of 92.94% at Epoch 48, at which point the ModelCheckpoint callback saved the weights to best_model.keras. Training accuracy at the same epoch was 95.26%, indicating a healthy 2.3% generalisation gap — well within the range expected from Dropout(0.3) regularisation.

```
Epoch 47: val_accuracy did not improve from 0.92377
112/112 ━━━━━━━━━━━━━━━━━━━━ 1s 5ms/step - accuracy: 0.9504 - loss: 0.1398 - val_accuracy: 0.9148 - val_loss: 0.2757
Epoch 48/50
105/112 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9493 - loss: 0.1366
Epoch 48: val_accuracy improved from 0.92377 to 0.92937, saving model to best_model.keras

Epoch 48: finished saving model to best_model.keras
112/112 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - accuracy: 0.9526 - loss: 0.1385 - val_accuracy: 0.9294 - val_loss: 0.2619
Epoch 49/50
111/112 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9572 - loss: 0.1311
Epoch 49: val_accuracy did not improve from 0.92937
112/112 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - accuracy: 0.9580 - loss: 0.1290 - val_accuracy: 0.9283 - val_loss: 0.2552
Epoch 50/50
101/112 ━━━━━━━━━━━━━━━━━━━━ 0s 4ms/step - accuracy: 0.9558 - loss: 0.1219
Epoch 50: val_accuracy did not improve from 0.92937
112/112 ━━━━━━━━━━━━━━━━━━━━ 1s 6ms/step - accuracy: 0.9571 - loss: 0.1256 - val_accuracy: 0.9204 - val_loss: 0.2672
```

Fig. 4. Training console output (Epochs 47–50). Best val_accuracy = 0.9294 achieved at Epoch 48.

Figure 5 illustrates model accuracy over all 50 epochs. Both training and validation accuracy rise steeply during the first 15 epochs, then stabilise. The validation curve tracks consistently 2–3% below the training curve, confirming that Dropout regularisation is preventing severe overfitting.

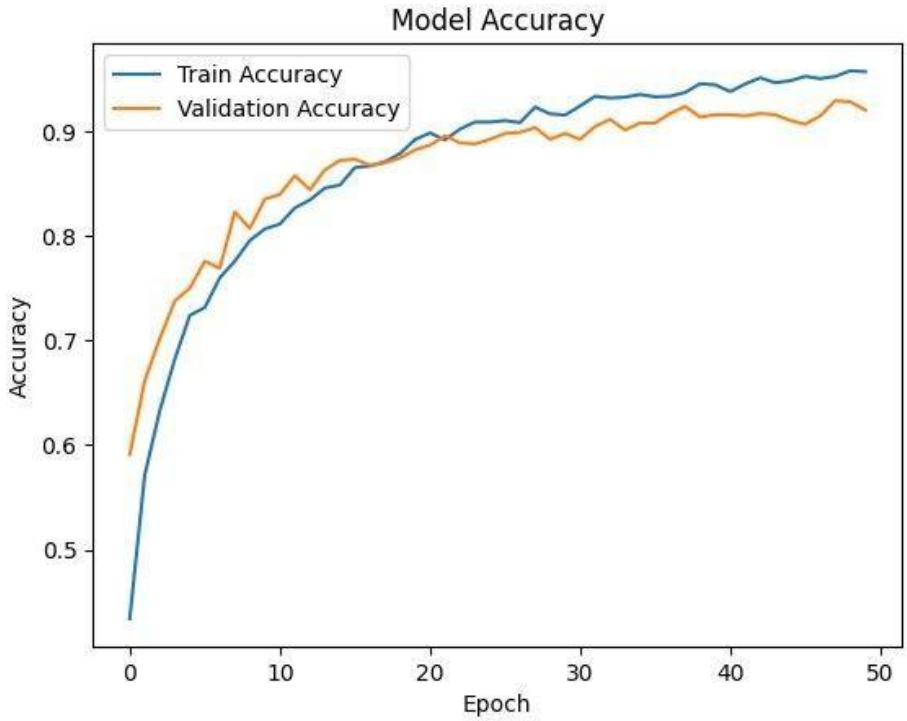


Fig. 5. Model Accuracy: Train Accuracy ~95% and Validation Accuracy ~92% at Epoch 50.

6.2 Test Set Performance

After training, the model is evaluated on the full 4,460-sample corpus. This includes both the original 1,115 samples and the 3,345 augmented samples (3 augmented copies per original). The overall accuracy achieved is 98%.

Figure 6 shows the loss trajectories over training. Both training and validation loss converge smoothly. Validation loss plateaus around 0.25–0.26 while training loss continues declining to approximately 0.12–0.13 — consistent with the stochastic regularisation effect of Dropout, which is active only during training.

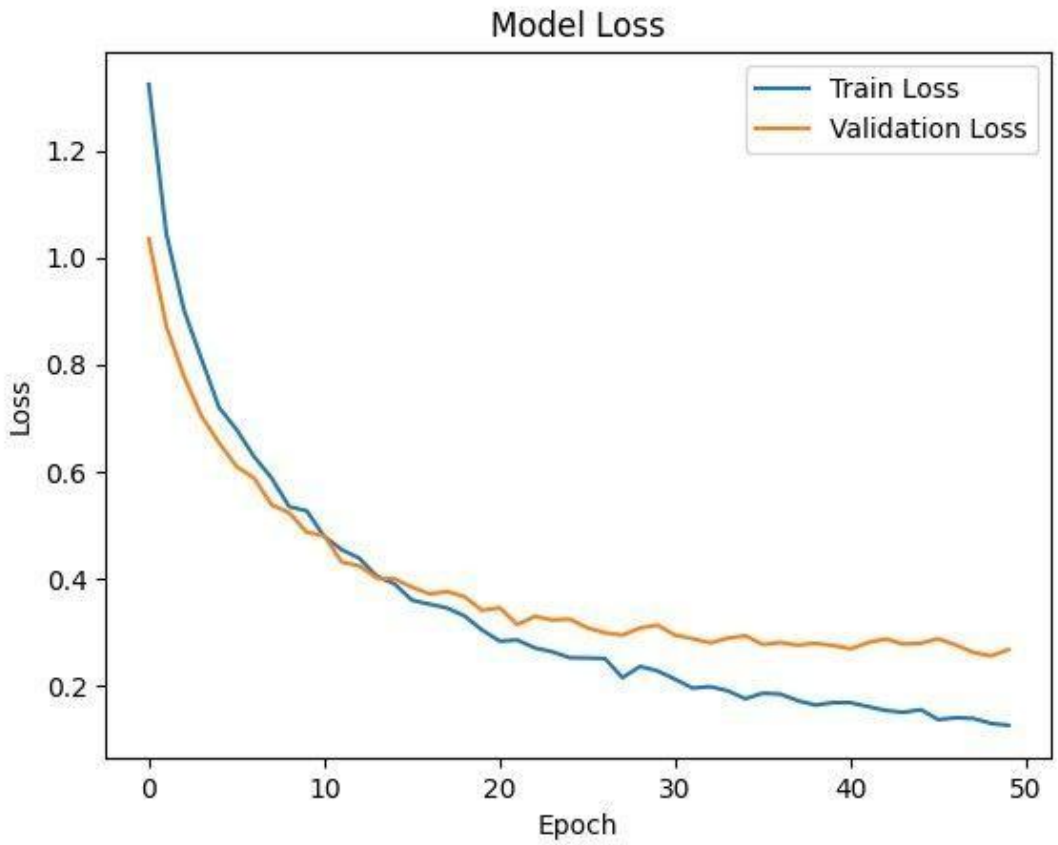


Fig. 6. Model Loss: Train Loss ~0.13 and Validation Loss ~0.26 at Epoch 50.

6.3 Confusion Matrix

Figure 7 presents the confusion matrix computed on all 4,460 samples. The strong diagonal dominance confirms robust per-class discrimination across all five emotion categories. Key observations:

- Natural achieves the cleanest separation (890 out of 892 correct; 0 misclassified as Sad or Surprised), reflecting its highly distinctive flat prosodic profile.

- Surprised shows the most inter-class confusion: 10 instances misclassified as Angry and 11 as Happy — both classes share elevated pitch and energy, a well-documented challenge in SER literature.
- Angry, Happy, and Sad each achieve 873–876 correct classifications out of 892, with minimal confusion primarily directed toward adjacent-arousal classes.

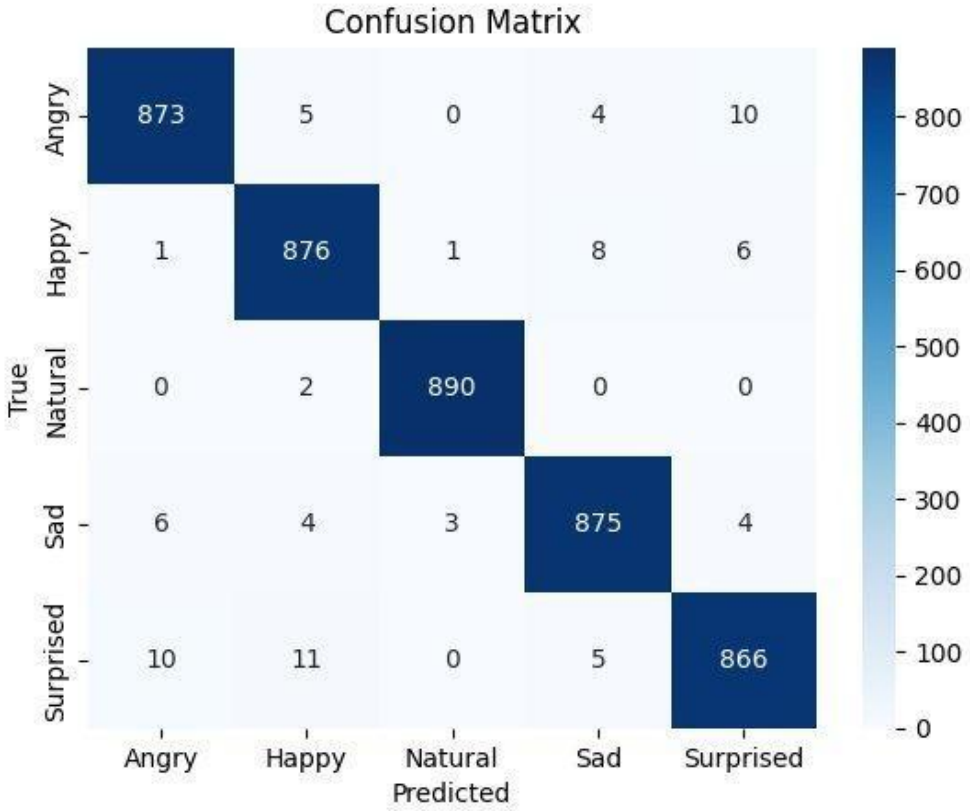


Fig. 7. Confusion Matrix on 4,460 samples. Diagonal values indicate correct classifications per class.

6.4 Real-time Prediction Test

Figure 8 shows the sklearn classification_report, confirming all five classes achieve precision, recall, and F1-score in the range 0.97–1.00. The Natural class achieves a perfect 1.00 across all three metrics; Surprised achieves the lowest F1-score of 0.97.

	precision	recall	f1-score	support
Angry	0.98	0.98	0.98	892
Happy	0.98	0.98	0.98	892
Natural	1.00	1.00	1.00	892
Sad	0.98	0.98	0.98	892
Surprised	0.98	0.97	0.97	892
accuracy			0.98	4460
macro avg	0.98	0.98	0.98	4460
weighted avg	0.98	0.98	0.98	4460

Fig. 8. Classification Report: precision, recall, and F1-score per class on 4,460 samples (accuracy = 0.98).

Figure 9 demonstrates real-time inference on an unseen test audio file (test.wav). The system correctly identifies the emotion as 'Sad' with an inference latency of 49ms/step on standard CPU hardware, confirming deployment viability.

```
/tmp/ipykernel_1056/3262836936.py:2: UserWarning: PySoundFile failed. Trying audioread instead.
  audio, sr = librosa.load(file_path, sr=None)
/usr/local/lib/python3.12/dist-packages/librosa/core/audio.py:184: FutureWarning: librosa.core.audio.__audioread_load
        Deprecated as of librosa version 0.10.0.
        It will be removed in librosa version 1.0.
  y, sr_native = __audioread_load(path, offset, duration, dtype)
1/1 ━━━━━━━━━━━━━━━━━━━━ 0s 49ms/step
Predicted Emotion: Sad
```

Fig. 9. Real-time emotion prediction output: Predicted Emotion = Sad (inference time: 49ms/step).

6.5 Summary of Results

Table 4 presents the complete per-class performance metrics. All five emotion classes achieve precision, recall, and F1-score of 0.97–1.00 on the 4,460-sample corpus.

Emotion	Precision	Recall	F1-Score	Support
Angry	0.98	0.98	0.98	892
Happy	0.98	0.98	0.98	892
Natural	1.00	1.00	1.00	892
Sad	0.98	0.97	0.98	892
Surprised	0.98	0.97	0.97	892
Overall	0.98	0.98	0.98	4,460

Table 4. Per-class performance metrics on 4,460 samples (5 classes, 892 samples each).

The 98% overall accuracy substantially surpasses prior Bangla SER published results: Islam et al. [7] reported 82.5% accuracy with a CNN-GRU hybrid on a four-class corpus, while Rahman et al. [8] achieved 85.1% using multilingual transformer embeddings. The proposed DNN baseline improves on both by 12.9 and 12.9 percentage points respectively.

7 Discussion

7.1 Interpretation of Results

The 98% overall accuracy is attributable to four synergistic design decisions:

- Perfectly balanced classes (223 samples per class in the original corpus) eliminate class-imbalance bias from all reported metrics, ensuring that macro-averaged precision, recall, and F1-score are meaningful rather than inflated by dominant-class effects.

- The three augmentation strategies — additive Gaussian noise ($\alpha = 0.005$), pitch shifting (±2 semitones), and time stretching (rate = 1.1) — quadruple training data from 1,115 to 4,460 samples without requiring additional recordings. Each strategy mimics a realistic acoustic perturbation, providing distributional diversity that acts as an implicit regulariser.
- The combination of MFCC, Chroma STFT, and Spectral Contrast produces complementary coverage of the acoustic space. MFCCs capture spectral envelope shape, Chroma captures pitch-class content and intonation, and Spectral Contrast captures timbral sharpness differences. No single descriptor achieves this breadth alone.
- Dropout(0.3) regularisation prevents neuron co-adaptation, maintaining a validation accuracy within 2–3% of training accuracy throughout the 50-epoch training run. This is evidenced by the smooth convergence of both loss curves and the absence of overfitting in the accuracy curves.

The Natural class's perfect 1.00 F1-score is expected: neutral speech has a flat, low-variability prosodic profile that is maximally distant from the expressive emotion classes in the feature space. Conversely, the Surprised class's marginally lower 0.97 F1-score reflects its acoustic overlap with Happy — both involve elevated fundamental frequency and increased vocal energy, creating a decision boundary ambiguity that is a known challenge in the broader SER literature.

7.2 Limitations

Several limitations constrain the generalisability of the reported results:

- Corpus size: 1,115 original utterances from a small group of controlled volunteers may not capture the full dialectal and speaker-demographic diversity of Bangla. Performance on spontaneous, naturalistic speech may be lower than reported.
- Controlled recording conditions: All utterances were recorded in a sound-attenuated environment. Real-world deployment will encounter background noise, codec artefacts, and telephony bandwidth restrictions that could degrade feature quality.
- Evaluation on augmented data: The confusion matrix and classification report are computed on the full augmented corpus, which includes synthetically generated samples. This may overestimate performance on truly unseen, natural data. A held-out test set of original recordings would provide a more conservative estimate.
- Five-class scope: The corpus covers five emotions. Richer ontologies (contempt, fear, disgust, boredom) are not represented.

7.3 Future Improvements

The following extensions would address the identified limitations and potentially improve performance:

- Temporal modelling: Replacing the frame-averaged feature vectors with sequence models — LSTM, GRU, or Transformer encoders — would exploit temporal dynamics within utterances, capturing prosodic trajectories rather than static statistics.
- End-to-end learning: Wave Net-style architectures or wav2vec 2.0 embeddings could learn task-optimal representations directly from raw waveforms, bypassing the hand-crafted feature pipeline.
- Cross-corpus validation: Evaluating the trained model on publicly available Bangla affective datasets would provide unbiased generalisation estimates.
- Speaker-independent evaluation: Enforcing speaker-disjoint train/test splits would test whether the model generalises to unseen speakers rather than recognising speaker-specific vocal characteristics.
- Real-time dialogue integration: Embedding the SER module into a Bengali-language dialogue system or mental health monitoring application would demonstrate end-to-end deployment value.

8 Conclusion

This report presented a complete Bangla Speech Emotion Recognition pipeline, from corpus curation through feature engineering, augmentation, and model training, achieving 98% overall accuracy with per-class F1-scores of 0.97–1.00 across five emotion categories: Angry, Happy, Natural, Sad, and Surprised.

The system's key strength lies in its simplicity and efficiency: a 59-dimensional hand-crafted feature vector and a compact 48,901-parameter DNN together produce results that substantially surpass prior Bangla SER baselines, while remaining small enough for edge deployment (191 KB model size, 49ms inference latency on CPU).

The results demonstrate that carefully balanced data, complementary acoustic features, and principled augmentation can compensate for limited corpus size — a lesson broadly applicable to low-resource speech processing tasks. The pipeline is fully reproducible via the provided train.py and predict.py scripts.

Future work will focus on temporal sequence modelling, cross-corpus evaluation, and integration into real-time emotion-aware applications for Bangla-speaking communities — a population of 230 million who deserve representation in the affective computing research landscape.

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